

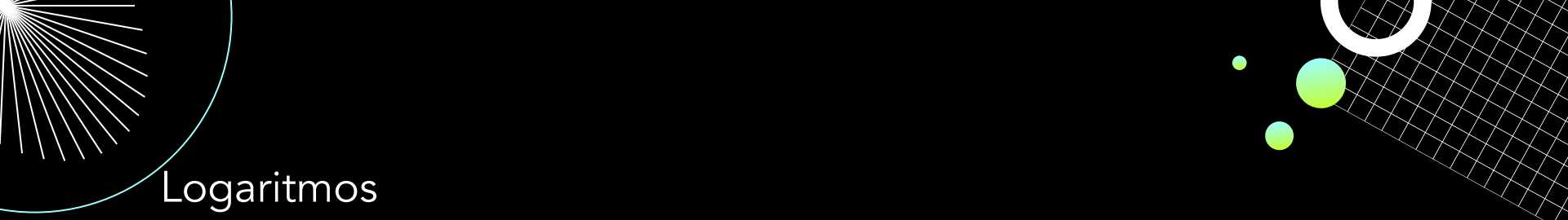


2ª SÉRIE

Matemática

Aula 04 - 2º Bim

Prof. Diogo Pelaes



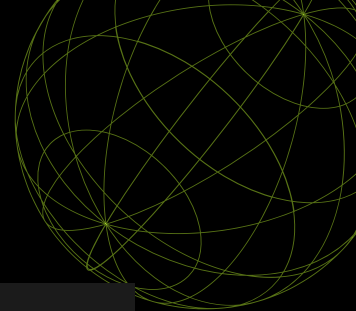
Logaritmos

- Pg. 41, Ex. 1 ao 12



IEZZI, Gelson; DEGENSZAJN, David; TAMARI, Marcio; PASMNIK, Guilherme. Identidade Saraiva: Matemática: Ensino Médio. 1. ed. São Paulo: Saraiva, 2026. Vol 2.

Logaritmos



Sendo a e b números reais e positivos, com $a \neq 1$, chama-se **logaritmo de b na base a** o expoente x ao qual se deve elevar a base a de modo que a potência a^x seja igual a b .

$$\log_a b = x \Leftrightarrow a^x = b$$

- a é a **base** do logaritmo;
- b é o **logaritmando**;
- x é o **logaritmo**.



- $\log_2 8 =$

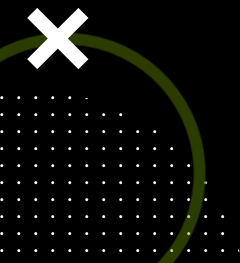
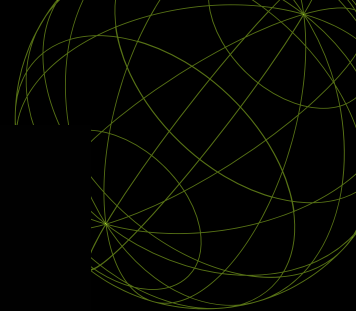
- $\log_4 1 =$

- $\log_2 \frac{1}{4} =$

- $\log_{\frac{1}{2}} 8 =$ |

- $\log_5 5 =$

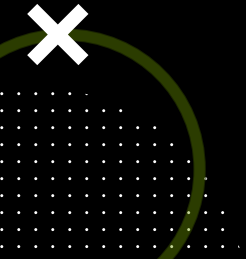
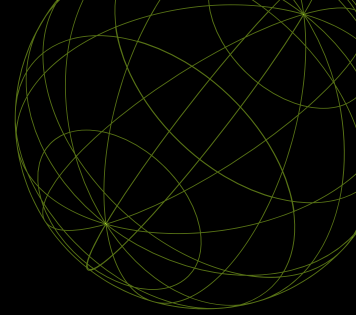
- $\log_{0,5} 0,25 =$

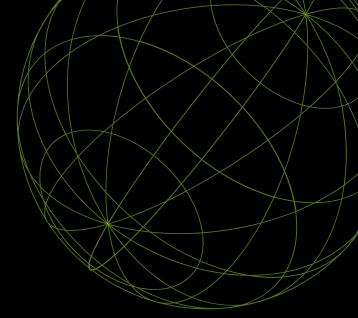


Exemplos

$$\log_{\sqrt[3]{9}} 3$$

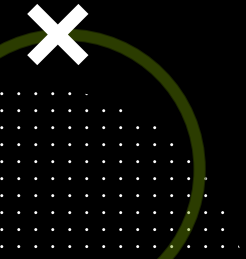
$$\log_{16} 0,25$$



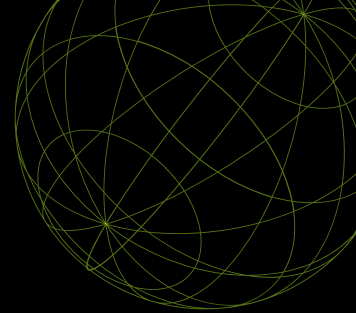


Exercício Resolvido

1. Qual é o número real x em $\log_x 4 = -2$?



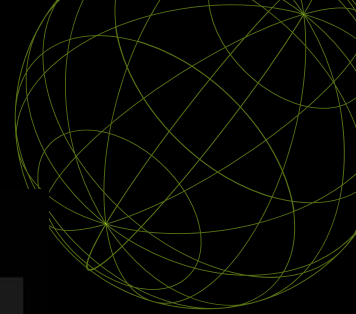
$$\log_{10} x = \log x$$



- $\log 10\,000 =$

- $\log \frac{1}{1000} =$





Consequências

- O logaritmo de 1 em qualquer base a é igual a 0.

$$\log_a 1 = 0, \text{ pois } a^0 = 1.$$

- O logaritmo de a na base a é igual a 1.

$$\log_a a = 1, \text{ pois } a^1 = a.$$

- A potência de base a e expoente $\log_a b$ é igual a b .

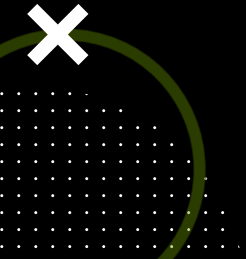
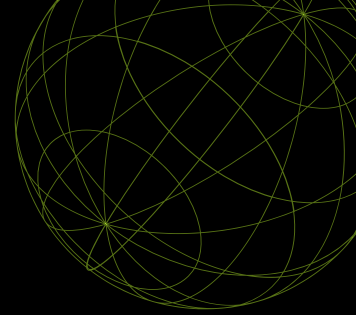
$$a^{\log_a b} = b.$$



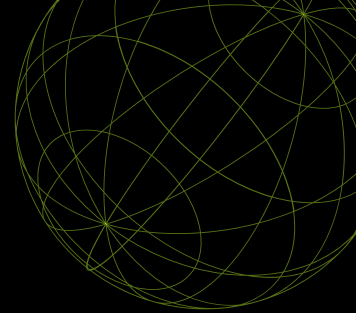
Exemplos

$$2^{\log_2 3}$$

$$5^{\log_5 4}$$

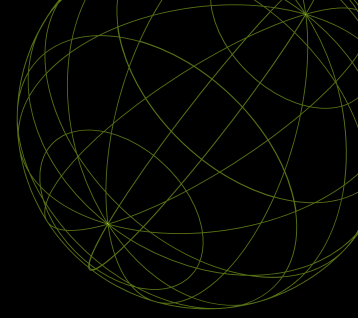


$$\log_a b = \log_a c \Leftrightarrow b = c$$



Qual é o número real x tal que $\log_5 (2x + 1) = \log_5 (x + 3)$?





Exercício Resolvido

2. Qual é o valor de $9^{\log_3 5}$?

